8.6 Consider the wage equation

$$WAGE_i = \beta_1 + \beta_2 EDUC_i + \beta_3 EXPER_i + \beta_4 METRO_i + e_i \quad \text{(XR8.6a)}$$

where wage is measured in dollars per hour, education and experience are in years, and $METRO = 1$ if the person lives in a metropolitan area. We have $N = 1000$ observations from 2013.

a. We are curious whether holding education, experience, and *METRO* constant, there is the same amount of random variation in wages for males and females. Suppose $\text{var}(e_i|\mathbf{x}_i, FEMALE = 0) = \sigma_M^2$ and $\text{var}(e_i|\mathbf{x}_i, FEMALE = 1) = \sigma_F^2$. We specifically wish to test the null hypothesis $\sigma_M^2 = \sigma_F^2$ against $\sigma_M^2 \neq \sigma_F^2$. Using 577 observations on males, we obtain the sum of squared OLS residuals, $SSE_M = 97161.9174$. The regression using data on females yields $\hat{\sigma}_F = 12.024$. Test the null hypothesis at the 5% level of significance. Clearly state the value of the test statistic and the rejection region, along with your conclusion.

(a) $H_0 : \sigma_M^2 = \sigma_F^2$ $\quad H_1 : \sigma_M^2 \neq \sigma_F^2$ $\quad WAGE_i = \beta_1 + \beta_2 EDUC_i + \beta_3 EXPER_i + \beta_4 METRO_i + e_i$, $k = 4$

男女工資變異性相等

(1) 男

$N_M = 577$, $SSE_M = 97161.9174$

$df = N_M - k = 577 - 4 = 573$

$\hat{\sigma}_M^2 = \frac{SSE_M}{df_M} = \frac{97161.9174}{573} \doteqdot 169.567$

(2) 女

$N_F = 1000 - 577 = 423$

$df = N_F - k = 423 - 4 = 419$

$\hat{\sigma}_F = 12.024 \Rightarrow \hat{\sigma}_F^2 = (12.024)^2 \doteqdot 144.5766$

$F = \frac{\hat{\sigma}_M^2}{\hat{\sigma}_F^2} = \frac{169.567}{144.5766} \doteqdot 1.1728 \sim F(573, 419)$.

RR : $F \geq F(0.975, 573, 419) = 1.196781$ or $F \leq F(0.025, 573, 419) = 0.837669$, $F = 1.1728 \notin RR$,

p-value $= 0.0818964 > \alpha = 0.05$, 不拒絕 H_0. #

b. We hypothesize that married individuals, relying on spousal support, can seek wider employment types and hence holding all else equal should have more variable wages. Suppose $\text{var}(e_i|\mathbf{x}_i, MARRIED = 0) = \sigma_{SINGLE}^2$ and $\text{var}(e_i|\mathbf{x}_i, MARRIED = 1) = \sigma_{MARRIED}^2$. Specify the null hypothesis $\sigma_{SINGLE}^2 = \sigma_{MARRIED}^2$ versus the alternative hypothesis $\sigma_{MARRIED}^2 > \sigma_{SINGLE}^2$. We add *FEMALE* to the wage equation as an explanatory variable, so that

右尾.

$$WAGE_i = \beta_1 + \beta_2 EDUC_i + \beta_3 EXPER_i + \beta_4 METRO_i + \beta_5 FEMALE + e_i \quad \text{(XR8.6b)}$$

Using $N = 400$ observations on single individuals, OLS estimation of (XR8.6b) yields a sum of squared residuals is 56231.0382. For the 600 married individuals, the sum of squared errors is 100,703.0471. Test the null hypothesis at the 5% level of significance. Clearly state the value of the test statistic and the rejection region, along with your conclusion.

(b) $H_0 : \sigma_{MARRIED}^2 = \sigma_{SINGLE}^2$ $\quad H_1 : \sigma_{MARRIED}^2 > \sigma_{SINGLE}^2$ $\quad k = 5$

(1) MARRIED

$N_M = 600$, $SSE_M = 100703.0471$

$df_M = N_M - k = 600 - 5 = 595$

$\hat{\sigma}_M^2 = \frac{SSE_M}{df_M} = \frac{100703.0471}{595} \doteqdot 169.2488$

(2) SINGLE

$N_S = 400$, $SSE_S = 56231.0382$

$df_S = N_S - k = 400 - 5 = 395$

$\hat{\sigma}_S^2 = \frac{SSE_S}{df_S} = \frac{56231.0382}{395} \doteqdot 142.3571$

$F = \frac{\hat{\sigma}_M^2}{\hat{\sigma}_F^2} = \frac{169.2488}{142.3571} \doteqdot 1.1889 \sim F(595, 395)$

RR : $F \geq F(0.95, 595, 395) = 1.16475$, $F \in RR$

p-value $= 0.03102054 < \alpha = 0.05$, 故 reject H_0. Married individuals have more variable wages. #

用方程式右邊的解釋變數作為「潛在影響變異數的候選變數」

c. Following the regression in part (b), we carry out the NR^2 test using the right-hand-side variables in (XR8.6b) as candidates related to the heteroskedasticity. The value of this statistic is 59.03. What do we conclude about heteroskedasticity, at the 5% level? Does this provide evidence about the issue discussed in part (b), whether the error variation is different for married and unmarried individuals? Explain.

用 EDUC、EXPER、METRO、FEMALE

輔助 regression：$\hat{e}_i^2 = \alpha_1 + \alpha_2 EDUC_i + \alpha_3 EXPER_i + \alpha_4 METRO_i + \alpha_5 FEMALE_i + v_i$

$H_0: \alpha_2 = \alpha_3 = \alpha_4 = \alpha_5 = 0$ H_1：至少一个 $\alpha \neq 0$.

$NR^2 = 59.03$，$df = 5-1 =$ z变数个数 $= 4$，$\chi^2_{(0.95,4)} \doteq 9.488$，p-value $= 4.637846 \times 10^{-12}$.

因 $59.03 > 9.488 \in RR$，故 reject H_0. 在 $\alpha = 0.05$ 下，存在异质变异数 #

在進行LM堅檢定時，我們設定用來解釋變異數的z變數，只有四個變數，婚姻狀態並未包含在內，因此無法解釋(b)小題

↳ EDUC、EXPER、METRO、FEMALE.

d. Following the regression in part (b) we carry out the White test for heteroskedasticity. The value of the test statistic is 78.82. What are the degrees of freedom of the test statistic? What is the 5% critical value for the test? What do you conclude?

原变数 4 个：EDUC、EXPER、METRO、FEMALE.

平方项 2 个：$EDUC^2$、$EXPER^2$ (Dummy variable 无须放入)

交乘项 6 个：EDUC × EXPER、EDUC × METRO、EDUC × FEMALE、EXPER × METRO、EXPER × FEMALE、METRO × FEMALE.

故 $df = 4+2+6 = 12$. (有 12 个 z 变数)

$\chi^2_{(0.95,12)} = 21.026$，

$\chi^2 = 78.82 > 21.026 \in RR$，p-value $= 6.926348 \times e^{-12}$，reject H_0. 在 $\alpha = 0.05$ 下，存在异质变异数. #

8.16 A sample of 200 Chicago households was taken to investigate how far American households tend to travel when they take a vacation. Consider the model

$df = 200-4 = 196$

c.I. $b_4 \pm t_c se(b_4)$

$$MILES = \beta_1 + \beta_2 INCOME + \beta_3 AGE + \beta_4 KIDS + e$$

MILES is miles driven per year, *INCOME* is measured in \$1000 units, *AGE* is the average age of the adult members of the household, and *KIDS* is the number of children.

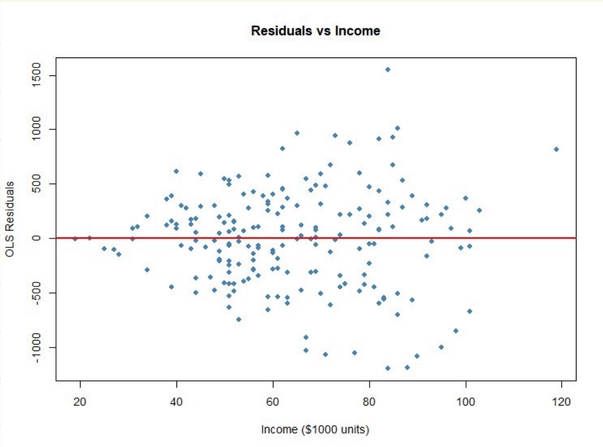
a. Use the data file *vacation* to estimate the model by OLS. Construct a 95% interval estimate for the effect of one more child on miles traveled, holding the two other variables constant.

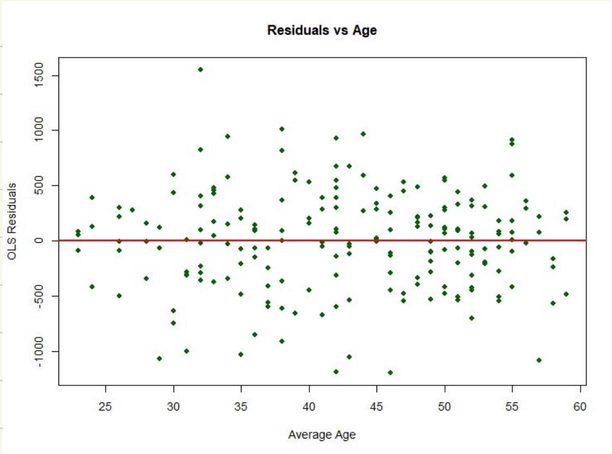
```
Call:
lm(formula = miles ~ income + age + kids, data = vacation)

Residuals:
     Min       1Q   Median       3Q      Max 
-1198.14  -295.31    17.98   287.54  1549.41 

Coefficients:
            Estimate Std. Error t value Pr(>|t|)    
(Intercept) -391.548    169.775  -2.306   0.0221 *  
income        14.201      1.800   7.889 2.10e-13 ***
age           15.741      3.757   4.189 4.23e-05 ***
kids         -81.826     27.130  -3.016   0.0029 ** 
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 452.3 on 196 degrees of freedom
Multiple R-squared:  0.3406,	Adjusted R-squared:  0.3305 
F-statistic: 33.75 on 3 and 196 DF,  p-value: < 2.2e-16
```

```
                 2.5 %    97.5 %
(Intercept) -726.36871 -56.72731
income        10.65097  17.75169
age            8.33086  23.15099
kids        -135.32981 -28.32302
```

Holding income and age constant, we are 95% confident that having one additional child decreases the annual vacation miles traveled by between 28.32 and 135.33 miles.

b. Plot the OLS residuals versus *INCOME* and *AGE*. Do you observe any patterns suggesting that heteroskedasticity is present?





The residual plots against INCOME and AGE both demonstrate a systematic 'fanning-out' pattern, where the dispersion of the OLS residuals increases significantly as the values of INCOME and AGE increase. This increasing variance in the residuals provides clear visual evidence of heteroskedasticity in the model.

c. Sort the data according to increasing magnitude of income. Estimate the model using the first 90 observations and again using the last 90 observations. Carry out the Goldfeld–Quandt test for heteroskedastic errors at the 5% level. State the null and alternative hypotheses.

$H_0: \sigma^2_{high} = \sigma^2_{low}$ $\quad H_1: \sigma^2_{high} > \sigma^2_{low}$

$\sigma^2_{high} = 315821.6$, $\sigma^2_{low} = 101744.6$, $F = \dfrac{\sigma^2_{high}}{\sigma^2_{low}} = \dfrac{315821.6}{101744.6} = 3.104061$

critical value = 1.428617 , 因 3.104061 > 1.428617 , reject Ho.

Because the calculated F-statistic (3.104) is greater than the critical value (1.429), we reject the null hypothesis and conclude that heteroskedasticity is present.

d. Estimate the model by OLS using heteroskedasticity robust standard errors. Construct a 95% interval estimate for the effect of one more child on miles traveled, holding the two other variables constant. How does this interval estimate compare to the one in (a)?

```
t test of coefficients:

             Estimate Std. Error t value  Pr(>|t|)
(Intercept) -391.5480   142.6548 -2.7447 0.0066190 **
income        14.2013     1.9389  7.3246 6.083e-12 ***
age           15.7409     3.9657  3.9692 0.0001011 ***
kids         -81.8264    29.1544 -2.8067 0.0055112 **
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
                  2.5 %     97.5 %
(Intercept) -672.883378 -110.21263
income        10.377633   18.02503
age            7.919934   23.56191
kids        -139.322973  -24.32986
```

Using heteroskedasticity robust standard errors, the new 95% confidence interval for the effect of one more child on miles traveled is [−139.32, −24.33]. Compared to the regular OLS interval in part (a) [−135.33, −28.32], this robust confidence interval is wider. The coefficient estimate remains exactly the same at −81.83, but the standard error has increased (from 27.13 to 29.15) to correct for the presence of heteroskedasticity.

e. Obtain GLS estimates assuming $\sigma_i^2 = \sigma^2 INCOME_i^2$. Using both conventional GLS and robust GLS standard errors, construct a 95% interval estimate for the effect of one more child on miles traveled, holding the two other variables constant. How do these interval estimates compare to the ones in (a) and (d)?

```
--- Conventional GLS 95% CI for kids ---
> print(ci_gls_regular["kids", ])
      2.5 %      97.5 %
-119.89450   -33.71808
```

```
--- Robust GLS 95% CI for kids ---
> print(ci_gls_robust["kids", ])
      2.5 %      97.5 %
-121.41339   -32.19919
```

Both GLS intervals ([−119.89, −33.72] and robust [−121.41, −32.20]) are narrower than the OLS intervals in (a) and (d), showing improved efficiency. Their similarity suggests our variance assumption successfully corrected the heteroskedasticity.